

RC Car Self-Driving Garage Checklist

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Print this page and use it while stripping the car down for the first self-driving build.

Keep Installed

- ☐ Raspberry Pi main computer
- ☐ Wide front USB webcam
- ☐ LiDAR
- ☐ Motor controller
- ☐ Drive motors
- ☐ Steering servo
- ☐ Main battery and power wiring
- ☐ Manual override controller or kill switch

Optional 1

- ☐ Hall sensor
- ☐ Front ultrasonic sensor

Keep these if you want extra safety and better speed or stop logic for version 1.

Optional 2

- ☐ Right ultrasonic sensor
- ☐ Pi camera using Picamera2
- ☐ GPS module
- ☐ IR sensors

Leave these out for the first build unless you have a specific reason to use them now.

Remove Now

- ☐ Headlights
- ☐ Brake lights
- ☐ Left indicator
- ☐ Right indicator
- ☐ Horn or buzzer
- ☐ MAX7219 LED display

- ☐ Voice assistant and speech features
- ☐ Photo capture extras
- ☐ Video recording extras
- ☐ Dashboard or UI extras not required for driving

Add Later

- ☐ Pi camera as rear or debug camera
- ☐ GPS for outdoor waypoint navigation
- ☐ Right ultrasonic for wall-follow behavior
- ☐ Lights or simple status LEDs
- ☐ Logging and video recording
- ☐ GUI and telemetry improvements

Build Order

1. ☐ Keep only the core drive, steering, compute, webcam, and LiDAR hardware.
2. ☐ Decide whether to keep hall sensor and front ultrasonic.
3. ☐ Remove lights, horn, display, and media or voice extras.
4. ☐ Verify manual driving still works.
5. ☐ Build version 1 self-driving using the wide webcam and LiDAR.
6. ☐ Add optional hardware only after the base system is stable.

Notes

- Version 1 should focus on simple, reliable self-driving.
- Do not add GPS or a second camera until the base system works.
- Keep a manual stop path available during every test.